

CS1231S TUTORIAL #3

Sets

Adapted from Prof Aaron Tan's slides

Learning objectives of this tutorial

Sets

- Roster, set-builder, replacement notation
- Membership, subset, set equality
- Power sets, Cartesian products
- Set operations
- Partition of a set
- Proving using set identities and element method

Sets

Set equality

$$A = B \Leftrightarrow \forall z (z \in A \Leftrightarrow z \in B).$$

Subset

$$A \subseteq B \Leftrightarrow \forall z (z \in A \Rightarrow z \in B).$$

Cartesian product

The Cartesian product of sets A and B ,

$$A \times B = \{(x, y) : x \in A \wedge y \in B\}.$$

Set operations

$$A \cup B = \{x : x \in A \vee x \in B\}.$$

$$A \cap B = \{x : x \in A \wedge x \in B\}.$$

$$A \setminus B = \{x : x \in A \wedge x \notin B\}.$$

Theorem 6.2.2. Set Identities

For all sets A, B, C in a context where U is the universal set.

Commutative Laws

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associative Laws

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Distributive Laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Identity Laws

$$A \cup \emptyset = A$$

$$A \cap U = A$$

Complement Laws

$$A \cup \bar{A} = U$$

$$A \cap \bar{A} = \emptyset$$

Double Complement Law

$$\overline{(\bar{A})} = A$$

Idempotent Laws

$$A \cup A = A$$

$$A \cap A = A$$

Universal Bound Laws

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

De Morgan's Laws

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

Absorption Laws

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

Complements of U and $\emptyset$

$$\bar{U} = \emptyset$$

$$\bar{\emptyset} = U$$

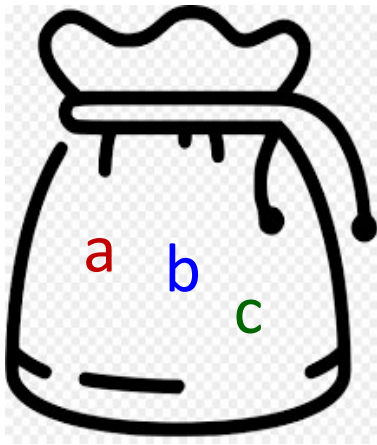
Set Difference Law

$$A \setminus B = A \cap \bar{B}$$

Theorem 6.2.1 Some Subset Relations

1. Inclusion of Intersection: For all sets A and B , (a) $A \cap B \subseteq A$ (b) $A \cap B \subseteq B$
2. Inclusion in Union: : For all sets A and B , (a) $A \subseteq A \cup B$ (b) $B \subseteq A \cup B$
3. Transitive property of subsets: For all sets A, B and C , $(A \subseteq B) \wedge (B \subseteq C) \rightarrow A \subseteq C$

Q1. (a) $\mathcal{P}(\{a, b, c\})$



$\mathcal{P}(\{a, b, c\}) =$

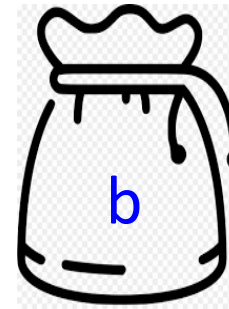
$\{ \emptyset, \{a\}, \{b\}, \{c\},$
 $\{a, b\}, \{a, c\}, \{b, c\},$
 $\{a, b, c\} \}$



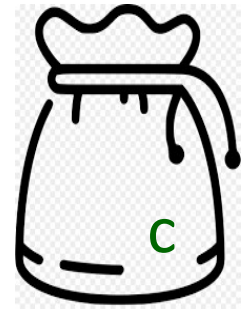
$\emptyset$



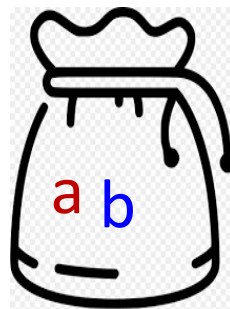
$\{a\}$



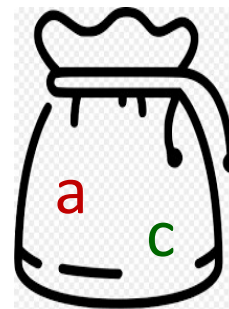
$\{b\}$



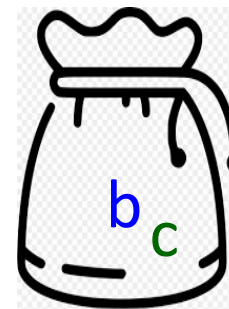
$\{c\}$



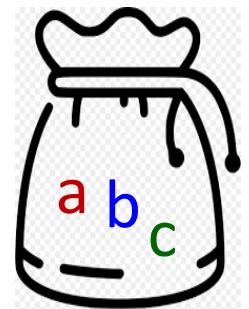
$\{a, b\}$



$\{a, c\}$



$\{b, c\}$



$\{a, b, c\}$

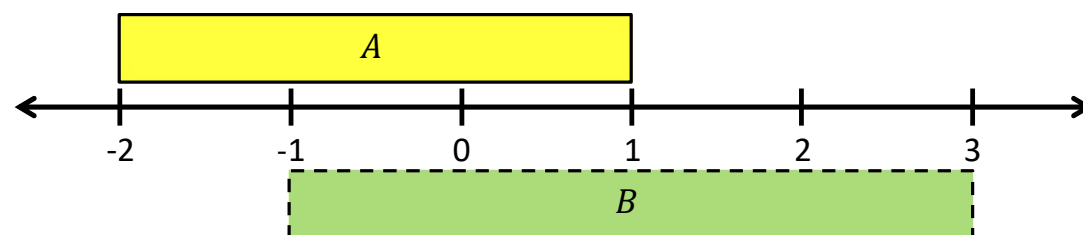
Q1. (b) $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$

$$\mathcal{P}(\emptyset) = \{\emptyset\}$$

$$\mathcal{P}(\mathcal{P}(\emptyset)) = \mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$$

$$\begin{aligned}\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset))) &= \mathcal{P}(\{\emptyset, \{\emptyset\}\}) \\ &= \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\end{aligned}$$

Q2. $A = \{x \in \mathbb{R} : -2 \leq x \leq 1\}$ and
 $B = \{x \in \mathbb{R} : -1 < x < 3\}$.



- (a) $A \cup B = \{x \in \mathbb{R} : -2 \leq x < 3\}$ or $[-2, 3)$
- (b) $A \cap B = \{x \in \mathbb{R} : -1 < x \leq 1\}$ or $(-1, 1]$
- (c) $\bar{A} = \{x \in \mathbb{R} : (x < -2) \vee (x > 1)\}$.
- (d) $\bar{A} \cap \bar{B} = \{x \in \mathbb{R} : (x < -2) \vee (x \geq 3)\}$.
- (e) $A \setminus B = \{x \in \mathbb{R} : -2 \leq x \leq -1\}$ or $[-2, -1]$

Q3. (AY2022/23 Sem2 mid-term test)

For each of the following statements, prove whether it is true or false.

- (a) There exist non-empty finite sets A and B such that $|A \cup B| = |A| + |B|$.

True Example?

Q3. (AY2022/23 Sem2 mid-term test)

For each of the following statements, prove whether it is true or false.

- (b) There exist non-empty finite sets A and B such that $|A \cup B| \neq |A| + |B|$.

True Example?

Q3. (AY2022/23 Sem2 mid-term test)

For each of the following statements, prove whether it is true or false.

(c) There exist finite sets A and B such that $A \times \mathcal{P}(B) = \mathcal{P}(A \times B)$.

1. Let $|A| = n, |B| = k$. False
2. Then $|\mathcal{P}(B)| = 2^k$ (cardinality of power set).
3. $|A \times B| = nk$ (cardinality of Cartesian product) and hence $|\mathcal{P}(A \times B)| = 2^{nk}$ (cardinality of power set).
4. Then $|A \times \mathcal{P}(B)| = n2^k$ (cardinality of Cartesian product).
5. Since $|A \times \mathcal{P}(B)| = n2^k \neq 2^{nk} = |\mathcal{P}(A \times B)|$, therefore the statement is false.

Can you write a correct proof for this?

Q3. (AY2022/23 Sem2 mid-term test)

For each of the following statements, prove whether it is true or false.

(c) There exist finite sets A and B such that $A \times \mathcal{P}(B) = \mathcal{P}(A \times B)$.

False

Proof:

1. Members of $A \times \mathcal{P}(B)$ are ordered pairs.
2. Members of $\mathcal{P}(A \times B)$ are sets. Note that $\mathcal{P}(A \times B)$ cannot be an empty set.
3. Ordered pairs and sets are different mathematical objects, and since $\mathcal{P}(A \times B)$ cannot be an empty set, hence $A \times \mathcal{P}(B) \neq \mathcal{P}(A \times B)$.

Why can't $\mathcal{P}(A \times B)$ be an empty set?



Q4. Let $A = \{2n + 1 : n \in \mathbb{Z}\}$ and $B = \{2n - 5 : n \in \mathbb{Z}\}$.

Is $A = B$? Yes or no?

$$A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A.$$

1. ($A \subseteq B$)

1.1. Let $a \in A$.

1.5. So $a \in B$.

2. ($B \subseteq A$)

2.1. Let $b \in B$.

2.5. So $b \in A$.

3. Hence $A = B$ by the definition of set equality.

Q4. Let $A = \{2n + 1 : n \in \mathbb{Z}\}$ and $B = \{2n - 5 : n \in \mathbb{Z}\}$.

Is $A = B$? Yes or no?

$$A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A.$$

1. ($A \subseteq B$)

1.1. Let $a \in A$.

1.2. Use the definition of A to find $n \in \mathbb{Z}$ such that $a = 2n + 1$.

1.3. Then $a = 2(n + 3) - 5$.

1.4. $n + 3 \in \mathbb{Z}$
(by closure of integers under +).

1.5. $\therefore a \in B$ (by the definition of B).

2. ($B \subseteq A$)

2.1. Let $b \in B$.

2.2. Use the definition of B to find $n \in \mathbb{Z}$ such that $b = 2n - 5$.

2.3. Then $b = 2(n - 3) + 1$.

2.4. $n - 3 \in \mathbb{Z}$
(by closure of integers under -).

2.5. $\therefore b \in A$ (by the definition of A).

3. $\therefore A = B$ (by the definition of set equality).

Q4. Let $A = \{2n + 1 : n \in \mathbb{Z}\}$ and $B = \{2n - 5 : n \in \mathbb{Z}\}$.
Is $A = B$?

$$A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A.$$

What about this?

1. $A = \{2n + 1 : n \in \mathbb{Z}\}$
2. $= \{2(n + 3) - 5 : n \in \mathbb{Z}\}$ (by basic algebra)
3. $= \{2k - 5 : k \in \mathbb{Z}\}$ (as $(n + 3) \in \mathbb{Z}$ by closure of integers under +)
4. $= B$.

It is rather unclear how elements of B can be expressed in the form of A .

When set equality is tested, we usually want to test that you know $A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A$.

Q5. Show that for all sets A, B, C ,
 $A \cap (B \setminus C) = (A \cap B) \setminus C$.

Definitions

$$A \cap B = \{x : x \in A \wedge x \in B\}.$$

$$A \setminus B = \{x : x \in A \wedge x \notin B\}.$$

1. $A \cap (B \setminus C) = \{x : x \in A \wedge x \in B \setminus C\}$ by the definition of $\cap$
2. $= \{x : x \in A \wedge (x \in B \wedge x \notin C)\}$ by the definition of $\setminus$
3. $= \{x : (x \in A \wedge x \in B) \wedge x \notin C\}$ by the associativity of $\wedge$
4. $= \{x : x \in A \cap B \wedge x \notin C\}$ by the definition of $\cap$
5. $= (A \cap B) \setminus C$ by the definition of $\setminus$



Break

Q6. Prove that for all sets A , B and C ,

$$A \setminus (B \setminus C) = (A \setminus B) \cup (A \cap C).$$

$$A \setminus (B \setminus C)$$

$$1. = A \setminus (B \cap \bar{C})$$

by the Set Difference Law;

$$2. = A \cap \overline{(B \cap \bar{C})}$$

by the Set Difference Law;

$$3. = A \cap (\bar{B} \cup \bar{\bar{C}})$$

by De Morgan's Law;

$$4. = A \cap (\bar{B} \cup C)$$

by the Double Complement Law;

$$5. = (A \cap \bar{B}) \cup (A \cap C)$$

by the Distributive Law;

$$6. = (A \setminus B) \cup (A \cap C)$$

by the Set Difference Law.

Set Difference Law: $A \setminus B = A \cap \bar{B}$;

De Morgan's Law: $\overline{A \cap B} = \bar{A} \cup \bar{B}$;

Double Complement Law: $\bar{\bar{A}} = A$;

Distributive Laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Q7(a). For all sets A and B , define
 $A \oplus B = (A \setminus B) \cup (B \setminus A)$.

Let $A = \{1,4,9,16\}$ and $B = \{2,4,6,8,10,12,14,16\}$.
Find $A \oplus B$.

$$A \setminus B = \{1,9\}.$$

$$B \setminus A = \{2,6,8,10,12,14\}.$$

$$A \oplus B = \{1,2,6,8,9,10,12,14\}.$$

Q7(b). For all sets A and B , define
 $A \oplus B = (A \setminus B) \cup (B \setminus A)$.

Set Difference Law:
 $A \setminus B = A \cap \bar{B}$.

Show that $A \oplus B = (A \cup B) \setminus (A \cap B)$ for all sets A, B .

- | | |
|---|-------------------------------|
| 1. $A \oplus B$ | |
| 2. $= (A \setminus B) \cup (B \setminus A)$ | by the definition of $\oplus$ |
| 3. $= (A \cap \bar{B}) \cup (B \cap \bar{A})$ | by the Set Difference Law |
| 4. $= ((A \cap \bar{B}) \cup B) \cap ((A \cap \bar{B}) \cup \bar{A})$ | by the Distributive Law |
| 5. $= (B \cup (A \cap \bar{B})) \cap (\bar{A} \cup (A \cap \bar{B}))$ | by the Commutative Law |
| 6. $= ((B \cup A) \cap (B \cup \bar{B})) \cap ((\bar{A} \cup A) \cap (\bar{A} \cup \bar{B}))$ | by the Distributive Law |
| 7. $= ((B \cup A) \cap (B \cup \bar{B})) \cap ((A \cup \bar{A}) \cap (\bar{A} \cup \bar{B}))$ | by the Commutative Law |
| 8. $= ((B \cup A) \cap U) \cap (U \cap (\bar{A} \cup \bar{B}))$ | by the Complement Law |
| 9. $= ((A \cup B) \cap U) \cap ((\bar{A} \cup \bar{B}) \cap U)$ | by the Commutative Law |
| 10. $= (A \cup B) \cap (\bar{A} \cup \bar{B})$ | by the Identity Law |
| 11. $= (A \cup B) \cap \overline{A \cap B}$ | by De Morgan's Law |
| 12. $= (A \cup B) \setminus (A \cap B)$ | by the Set Difference Law |

Q8. Let A, B be sets.

Show that $A \subseteq B$ if and only if $A \cup B = B$.



Definition.

$$A \cup B = \{x : x \in A \vee x \in B\}.$$

1. ("Only if" / $\Rightarrow$)

To show $A \cup B = B$,
need to show $A \cup B \subseteq B$
and $A \cup B \supseteq B$.

1.1. Suppose $A \subseteq B$.

1.2. (To show $A \cup B \subseteq B$.)

1.2.1. Let $z \in A \cup B$.

1.2.2. Then $z \in A$ or $z \in B$ by defn of $\cup$.

1.2.3. Case 1: Suppose $z \in A$.

Then $z \in B$ as $A \subseteq B$ by 1.1.

1.2.4. Case 2: Suppose $z \in B$.

Then $z \in B$.

1.2.5. In all cases, $z \in B$.

2. ("If" / $\Leftarrow$) ...

1.3. (To show $A \cup B \supseteq B$.)

1.3.1. Let $z \in B$.

1.3.2. Then $z \in A \vee z \in B$ by generalization.

1.3.3. So $z \in A \cup B$ by definition of $\cup$.

1.4. $\therefore A \cup B = B$ by definition of set equality.

Definitions.

$$A \subseteq B \Leftrightarrow \forall z(z \in A \Rightarrow z \in B).$$

$$A = B \Leftrightarrow \forall z(z \in A \Leftrightarrow z \in B).$$

Q8. Let A, B be sets.

Show that $A \subseteq B$ if and only if $A \cup B = B$.



Definition.

$$A \cup B = \{x : x \in A \vee x \in B\}.$$

1. ("Only if" / $\Rightarrow$) ...

2. ("If" / $\Leftarrow$)

2.1. Suppose $A \cup B = B$.

2.2. Let $z \in A$.

2.2.1. Then $z \in A \vee z \in B$

by generalization.

2.2.2. So $z \in A \cup B$

by definition of $\cup$.

2.2.3. So $z \in B$

as $A \cup B = B$ by 2.1.

2.3. Therefore, $A \subseteq B$.

3. $\therefore A \subseteq B$ if and only if $A \cup B = B$ by lines 1 & 2.

Definitions.

$$A \subseteq B \Leftrightarrow \forall z(z \in A \Rightarrow z \in B).$$

$$A = B \Leftrightarrow \forall z(z \in A \Leftrightarrow z \in B).$$

Q8. Let A, B be sets.

Show that $A \subseteq B$ if and only if $A \cup B = B$.

Definitions.

$$A \subseteq B \iff \forall z(z \in A \Rightarrow z \in B).$$

$$A = B \iff \forall z(z \in A \Leftrightarrow z \in B).$$

(General strategy)

To show that $A \subseteq B$:

- Let $z \in A$
- ...
- $z \in B$
- Hence, $A \subseteq B$



Q9. For all sets A, B, C , $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

(a) Aiken wrote this proof:

What's wrong with the proof?
(at least four mistakes)

1. $x \in (A \setminus B) \cup (B \setminus A)$
 2. $\Rightarrow x \in (A \setminus B)$ or $x \in (B \setminus A)$
 3. $\Rightarrow x \in A$ and $x \notin B$ or $x \in B$ and $x \notin A$
 4. $\Rightarrow x \in A$ or $x \in B$ and $x \notin A$ and $x \notin B$
 5. $\Rightarrow x \in (A \cup B)$ and $x \in \overline{(A \cap B)}$
 6. $\Rightarrow x \in (A \cup B) \cap \overline{(A \cap B)}$
 7. $\Rightarrow x \in (A \cup B) \setminus (A \cap B)$.
- Therefore, $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

Q9. For all sets A, B, C , $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

(b) Prove or disprove the claim.

1. $(A \setminus B) \cup (B \setminus A)$
2. $= (A \cap \bar{B}) \cup (B \cap \bar{A})$ by the definition of $\setminus$
3. $= ((A \cap \bar{B}) \cup B) \cap ((A \cap \bar{B}) \cup \bar{A})$ by the distributive law
4. $= (B \cup (A \cap \bar{B})) \cap (\bar{A} \cup (A \cap \bar{B}))$ by the commutative law (x2)
5. $= ((B \cup A) \cap (B \cup \bar{B})) \cap ((\bar{A} \cup A) \cap (\bar{A} \cup \bar{B}))$ by the distributive law (x2)
6. $= ((B \cup A) \cap U) \cap ((\bar{A} \cup A) \cap (\bar{A} \cup \bar{B}))$ by the complement law
7. $= (B \cup A) \cap ((\bar{A} \cup A) \cap (\bar{A} \cup \bar{B}))$ by the identity law
8. $= (A \cup B) \cap ((\bar{A} \cup \bar{B}) \cap (A \cup \bar{A}))$ by the commutative law (x3)
9. $= (A \cup B) \cap ((\bar{A} \cup \bar{B}) \cap U)$ by the complement law
10. $= (A \cup B) \cap (\bar{A} \cup \bar{B})$ by the identity law
11. $= (A \cup B) \cap \overline{(A \cap B)}$ by De Morgan's law
12. $= (A \cup B) \setminus (A \cap B)$ by the definition of $\setminus$

Q9. For all sets A, B, C , $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

(b) Prove or disprove the claim.

7. For sets A and B , define $A \oplus B = (A \setminus B) \cup (B \setminus A)$.

a. Let $A = \{1, 4, 9, 16\}$ and $B = \{2, 4, 6, 8, 10, 12, 14, 16\}$. Find $A \oplus B$.

b. Using **set identities** (Theorem 6.2.2), prove that for all sets A and B ,

$$A \oplus B = (A \cup B) \setminus (A \cap B).$$

Q9. For all sets A, B, C , $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

(b) Prove or disprove the claim.

1. Let $A \oplus B = (A \setminus B) \cup (B \setminus A)$, as defined in Tutorial 3 Q7
2. $A \oplus B = (A \cup B) \setminus (A \cap B)$ for all sets A, B (by Tutorial 3 Q7b)
3. Hence, $(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$

Q10. $HSWW$ is the set of students in the Hogwarts School of Witchcraft and Wizardry, and G, H, R, S are the sets of students in the 4 houses.
What are the necessary conditions for $\{G, H, R, S\}$ to be a partition of $HSWW$?



Every student is in exactly one of the four houses.
Each house has at least one student.

$$G \cap H = G \cap R = G \cap S = H \cap R = H \cap S = R \cap S = \emptyset$$

$$G \cup H \cup R \cup S = HSWW$$

$$G \neq \emptyset \wedge R \neq \emptyset \wedge H \neq \emptyset \wedge S \neq \emptyset$$

Q11-12. For sets $A_m, A_{m+1}, \dots, A_n$, define

$$\bigcup_{i=m}^n A_i = A_m \cup A_{m+1} \cup \dots \cup A_n$$

and

$$\bigcap_{i=m}^n A_i = A_m \cap A_{m+1} \cap \dots \cap A_n$$



Q11. Let $A_i = \{x \in \mathbb{Z} : i \leq x \leq 2i\}$ for all integers i .

Write A_{-2} , $\bigcup_{i=3}^5 A_i$ and $\bigcap_{i=3}^5 A_i$ in set-roster notation, set-builder notation and interval notation.

$$A_{-2} = \emptyset$$

$$\bigcup_{i=3}^5 A_i = \{3,4,5,6,7,8,9,10\} = \{x \in \mathbb{Z} : 3 \leq x \leq 10\} = [3,10]$$

$$\bigcap_{i=3}^5 A_i = \{5,6\} = \{x \in \mathbb{Z} : 5 \leq x \leq 6\} = [5,6]$$

Q12. Let $V_i = \left\{x \in \mathbb{R} : -\frac{1}{i} \leq x \leq \frac{1}{i}\right\} = \left[-\frac{1}{i}, \frac{1}{i}\right]$ for all positive integers i .

(a) $\bigcup_{i=1}^4 V_i = [-1, 1]$

(b) $\bigcap_{i=1}^4 V_i = \left[-\frac{1}{4}, \frac{1}{4}\right]$

(c) $\bigcup_{i=1}^n V_i = [-1, 1]$

(d) $\bigcap_{i=1}^n V_i = \left[-\frac{1}{n}, \frac{1}{n}\right]$

(e) Are $V_1, V_2, V_3, \dots$ mutually disjoint?
Why?

THE END